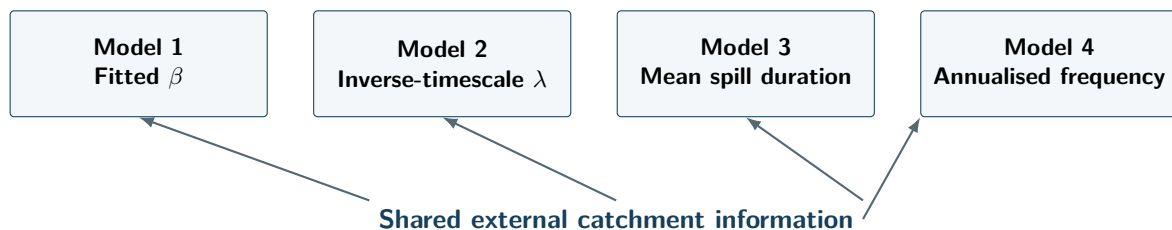


New Machine Learning Models 1–4 Predicting CSO Outcomes at Unseen WWTW Catchments

A compact research report explaining what each model predicts, how to read the diagnostics, what the results actually mean, and which findings matter most.

Core conclusion. The four models contain genuine but limited out-of-sample information. Point estimates beat the naive mean baseline in every target, but the gains are small: 2.6%–7.0%, with test R^2 only 0.052–0.133. Annualised spill frequency is the strongest target. Inverse-timescale λ is the weakest and most unstable. The dominant limitation is not a lack of algorithm complexity; it is missing hydraulic/operational state and imperfect target information.



*70% training | 15% validation | 15% locked test | RMSE headline metric
The locked test is used only after model selection.*

Executive interpretation: what the four models actually say

Target	Selected model	Test RMSE	Gain vs dummy	R^2	Interpretation
Fitted β	OLS, inter- pretable static	0.653	4.9%	0.073	Small signal; improvement uncertainty includes zero.
Inverse-timescale λ	OLS, inter- pretable static	3.855	4.9%	0.095	Weak / inconclusive; severe extreme-value instability.
Mean duration	Ridge, full static + dynamic	1.631	2.6%	0.052	Smallest baseline gain; company information matters strongly.
Annualised frequency	Ridge, full static + dynamic	1.341	7.0%	0.133	Strongest of the four, but still only modest predictive skill.

The result in plain English. These models can tell us something about which unseen WWTWs are likely to have higher or lower outcomes, but they cannot yet make accurate site-level predictions across the full observed range. Predictions cluster around typical values and struggle with extreme sites. The frequency model has the clearest evidence of added information beyond a constant baseline; the other three gains remain qualified by uncertainty.

Five findings to tell a supervisor first

1. **Frequency is the most predictable outcome.** It improves RMSE by 7.0% relative to the dummy baseline, has the highest test R^2 (0.133), and its bootstrap improvement interval is entirely positive: [0.8%, 12.1%].
2. **Duration is not rescued by having the largest sample.** Despite $n = 1,228$, the gain is only 2.6% and $R^2 = 0.052$. This directly supports the report's learning-curve conclusion that richer information matters more than simply adding similar WWTWs.
3. **λ is the weakest target.** Its 4.9% point improvement has a bootstrap interval spanning zero and the locked-test plot is dominated by extreme λ values that the model cannot reproduce.
4. **Annual dynamic rainfall adds only incremental skill.** It lowers test RMSE for β , duration and frequency, with the largest reduction for frequency (0.054), but it worsens λ by 0.101.
5. **The main problem is missing system state, not model simplicity.** All learning curves approach plateaus and the report identifies missing storage, sewer topology, pumping state, hydraulic headroom, groundwater/soil wetness and event-specific antecedent conditions.

Critical comparison rule. Never compare the raw RMSE numbers between β , λ , duration and frequency. They are different targets on different transformed scales. Cross-model comparison should use improvement over each target's own dummy baseline, R^2 , robustness, and the qualitative diagnostics.

1. How to read the metrics and plots

RMSE: the main prediction-error metric

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}$$

Lower is better. RMSE penalises large errors more heavily because errors are squared before averaging. In this programme the headline RMSE is calculated on the natural-log target scale, because every outcome is positive and strongly skewed.

How to interpret the dummy comparison. The dummy model predicts a constant training-set mean. The reported gain is

$$100 \times \frac{\text{RMSE}_{\text{dummy}} - \text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{dummy}}}.$$

A 7% gain does *not* mean the model is 93% accurate. It means its RMSE is 7% lower than a very simple constant predictor on the same target and test set.

R^2 : how much test-set variation is captured

$R^2 = 1$ is perfect. $R^2 = 0$ is no better than predicting the test-set mean in squared-error terms. Here the values 0.052–0.133 are low, so most between-WWTW variation remains unexplained by the measured feature bank.

Spearman correlation: ranking rather than exact prediction

Spearman correlation asks whether higher observed outcomes tend to receive higher predictions. Values of 0.252–0.346 show only weak-to-moderate ranking information. A model can rank sites somewhat correctly while still making poor numerical predictions.

Bootstrap improvement interval: is the baseline gain stable?

The report resamples the fixed locked-test predictions 2,000 times. If the interval crosses 0%, the measured improvement could plausibly disappear under test-set resampling. This happens for β , λ and duration. Frequency is the only target whose interval stays above zero.

Observed-versus-predicted plot

The dashed diagonal is perfect prediction. Points lying on the line are exact. A cloud compressed horizontally or vertically toward the centre means the model is shrinking predictions toward typical values and failing to reproduce extremes.

Residual plots

Residuals should ideally be centred around zero without obvious structure. Large asymmetric tails or extreme points show sites where the model misses badly. In this report, difficult extremes are retained rather than removed, so the residual diagnostics expose genuine limitations.

2. Experimental design: why the locked test matters

The central methodological strength. The 15% test set is not used to choose the algorithm or feature configuration. Candidate selection happens on validation data; only then is the model frozen, refitted on train+validation (85%), and evaluated once on the locked 15% test set.



Separate cohorts, not one forced complete-case dataset

Each target uses every WWTW with a valid outcome. This preserves 819 β sites, 783 λ sites, 1,228 duration sites and 967 frequency sites. No site is discarded merely because another outcome is missing.

Candidate models

Method	What it does in this programme
OLS	Multiple linear regression. Direct, interpretable coefficients with no regularisation.
Ridge	Linear regression with an L_2 penalty; useful when predictors are correlated because coefficients are shrunk and stabilised.
Elastic Net	Linear regression with both L_1 and L_2 penalties; can shrink some coefficients strongly while handling correlation.
Dummy mean	No catchment information. Predicts a constant and defines the baseline to beat.

Predictor bank

Family	Predictors included
Annual dynamic rain-fall	PRCPTOT, Rx1day, Rx5day, SDII, R10 days, R20 days, consecutive wet days, consecutive dry days.
Hydrogeology	High-productivity, low-productivity, essentially no groundwater, intergranular-flow geology.
Urban / demographic	Continuous developed-land index; population density.
Static rainfall	Mean annual rainfall; mean winter rainfall.
System context	Pre-1973 building share; treatment capacity utilisation; WWTW catchment area; mean terrain slope.
Sensitivity-only	Water company identity. Coordinates are used for spatial grouping, not prediction.

Moderate-productivity geology and fracture flow are reference categories to avoid exact compositional singularity. All preprocessing is learned inside scikit-learn pipelines from the relevant fitting data only.

3. Model 1: predicting fitted β

Question. Can external catchment conditions predict the fitted long-duration tail-shape parameter β at an unseen WWTW?

Answer. Only modestly. There is some out-of-sample signal, but the improvement over the constant baseline is small and its bootstrap interval crosses zero.

Data	819 total: 573 train, 123 validation, 123 test	Selected model	OLS, interpretable static
Test RMSE	0.653	Dummy RMSE	0.687
Gain	4.9%	Test R^2	0.073
Spearman	0.257	Bootstrap gain	[-1.0%, 10.3%]

What this means

- The model reduces error by only 4.9% relative to predicting one constant value for all unseen WWTWs.
- $R^2 = 0.073$ means the measured features capture only a small fraction of test-set variation in β .
- The bootstrap interval includes 0%, so the apparent improvement is not stable enough to call a strong result.
- The selected model is simple OLS rather than Ridge/Elastic Net. With this feature bank and validation split, extra regularisation/feature complexity did not produce a clear validation advantage.

What information did the model rely on?

Population density, intergranular-flow geology and low-productivity geology are the most relied-on measured features. Annual dynamic rainfall lowers test RMSE by 0.015 in the pre-specified sensitivity comparison. Adding company identity changes RMSE by +0.002, effectively no useful held-out improvement.

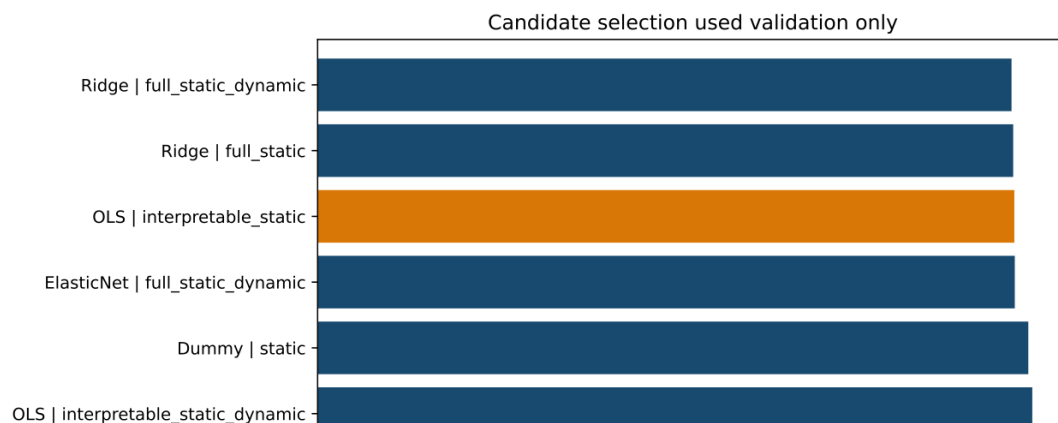


Figure 1: Validation comparison for Model 1. Candidate selection was performed on validation data only; the highlighted OLS interpretable-static specification was frozen before the locked test.

Locked-test diagnostics: where Model 1 succeeds and fails

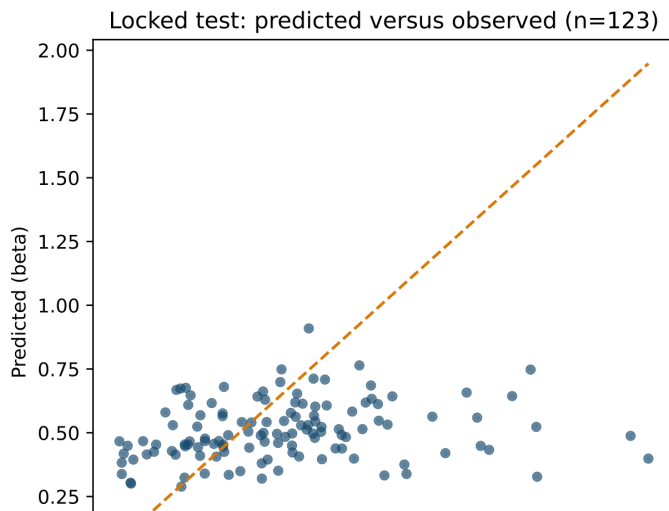


Figure 2: Observed versus predicted β . The dashed diagonal is perfect prediction.

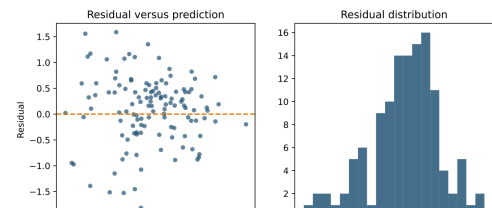


Figure 3: Residual diagnostics on the transformed scale.

How to read these diagnostics. Predictions occupy a narrow central band while observed β extends much further upward. The model therefore recognises some ordering between sites, but it does not reproduce many high- β observations. Residual spread remains substantial around zero. This is the visual form of a low- R^2 model that pulls difficult sites toward typical values.

Core finding for Model 1. External catchment structure contains a real-looking but weak β signal. The model is useful as evidence that β is not completely unpredictable, but it is not accurate enough for confident site-level prediction. The learning curve is already near a plateau, so improved β target quality and richer process information are more promising than simply adding similar WWTWs.

4. Model 2: predicting inverse-timescale λ

Question. Can external conditions predict fitted inverse-timescale λ at an unseen WWTW?

Answer. The evidence is weak and inconclusive. λ is the least trustworthy predictive target in the four-model programme because extreme fitted values dominate errors and the baseline gain is not robust.

Data	783 total: 547 train, 118 validation, 118 test	Selected model	OLS, interpretable static
Test RMSE	3.855	Dummy RMSE	4.056
Gain	4.9%	Test R^2	0.095
Spearman	0.264	Bootstrap gain	[-2.3%, 8.6%]

What this means

- The point estimate beats the dummy by 4.9%, but the bootstrap interval extends below 0%. The improvement may vanish under plausible test-set resampling.
- $R^2 = 0.095$ is still low, and the physical-scale predictions are dominated by an extremely right-skewed λ distribution.
- Dynamic annual rainfall *worsens* test RMSE by 0.101. Company identity also worsens it by 0.014. Neither adds reliable held-out value here.
- Population density, low-productivity geology and intergranular-flow geology are the most relied-on measured features.

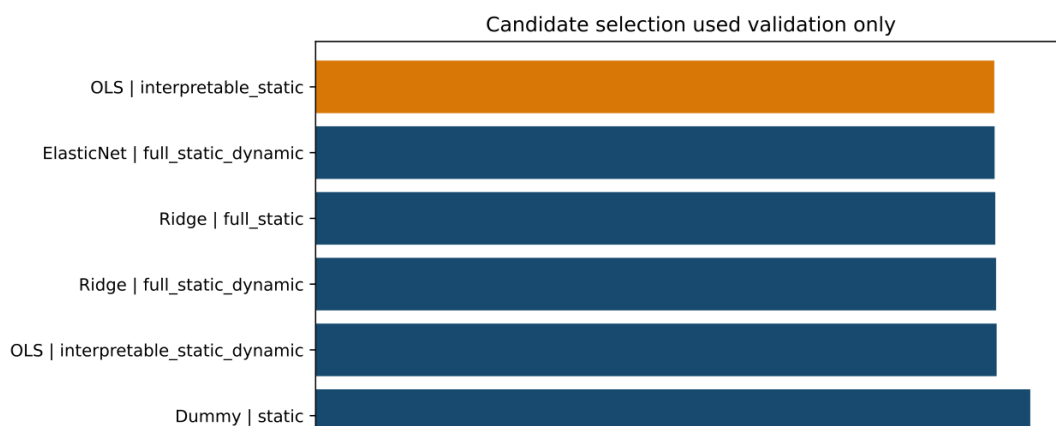


Figure 4: Validation model comparison for Model 2. OLS with the interpretable-static feature set was frozen before test evaluation.

Locked-test diagnostics: why λ is problematic

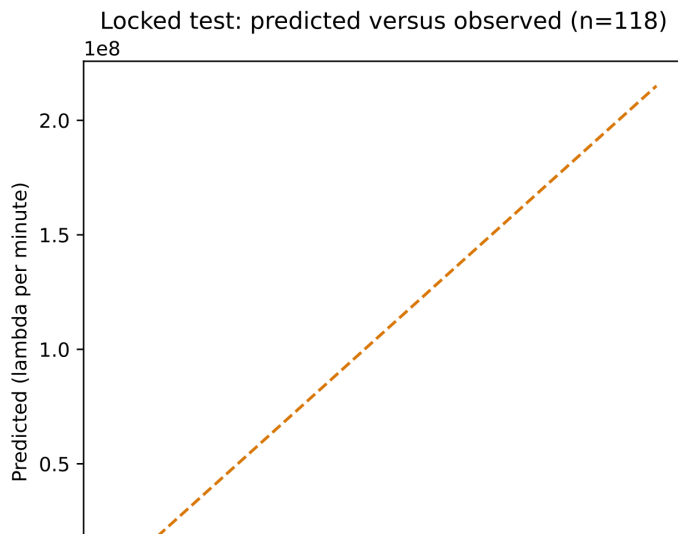


Figure 5: Observed versus predicted λ in original units.

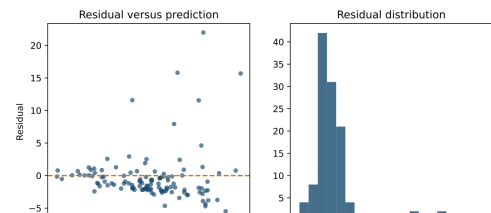


Figure 6: Residual diagnostics; large transformed-scale outliers remain.

Interpretation. A small number of fitted λ values are enormously larger than the rest. The model predicts values near the main cloud and therefore misses these extremes by orders of magnitude. The residual distribution retains several very large positive errors. This directly demonstrates why original-unit error is unstable and why λ receives the report’s “weak / inconclusive” verdict.

Core finding for Model 2. The current external predictors do not support reliable λ prediction. The report identifies fitted-parameter uncertainty and joint β – λ identifiability as target-quality problems. Improving the tail fit, retaining parameter covariance and adding event-mixture/hydraulic descriptors should take priority over trying more aggressive algorithms.

5. Model 3: predicting mean spill duration

Question. Can the measured environmental and catchment variables predict mean spill duration at an unseen WWTW?

Answer. Only modestly. This model has the largest dataset but the smallest improvement over the dummy baseline.

Data	1,228 total: 858 train, 185 validation, 185 test	Selected model	Ridge, full static + dynamic
Test RMSE	1.631	Dummy RMSE	1.675
Gain	2.6%	Test R^2	0.052
Spearman	0.252	Bootstrap gain	[-1.5%, 6.5%]

What this means

- A 2.6% reduction in RMSE is the smallest point gain among the four models.
- $R^2 = 0.052$ shows that very little site-to-site duration variation is captured by the current external variables.
- The bootstrap interval crosses zero, so the baseline improvement is qualified rather than definitive.
- Ridge is selected because the full static + dynamic feature bank contains correlated information; shrinkage stabilises the linear predictor rather than creating a nonlinear model.

The most important sensitivity result for duration

Annual dynamic rainfall reduces test RMSE by only 0.011, but adding company identity reduces RMSE by 0.131. This is much larger than the company effect for the other targets. Within this report, that is strong evidence that operator/regional information is proxying important duration-relevant factors that the measured feature bank does not contain.

Static mean winter rainfall, static mean annual rainfall and annual SDII are the most relied-on measured features.

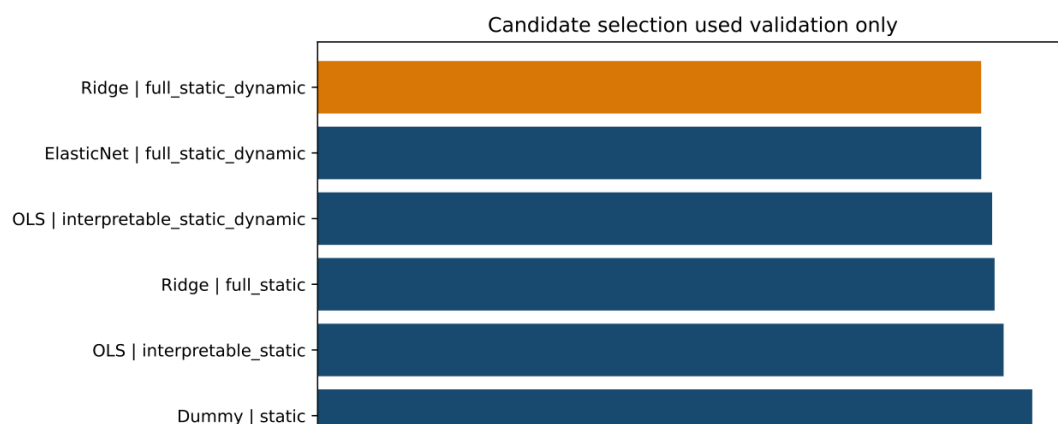


Figure 7: Validation comparison for Model 3. Ridge with the full static + annual-dynamic feature bank is selected.

Locked-test diagnostics: central values are easier than extremes

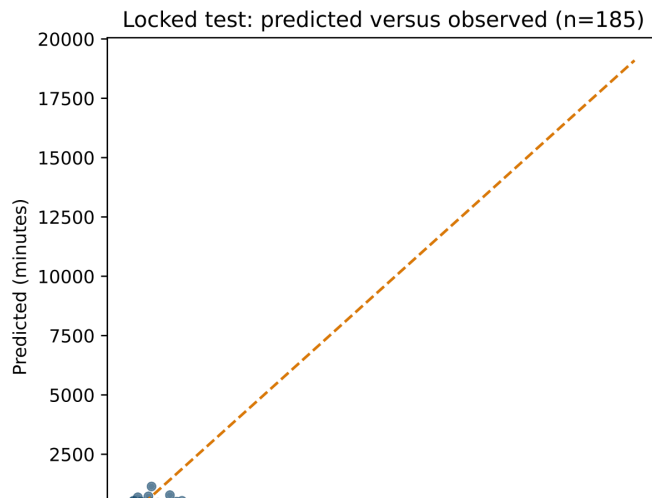


Figure 8: Observed versus predicted mean duration in minutes.

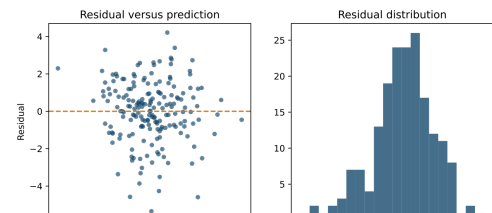


Figure 9: Residual diagnostics on the log scale.

How to read these diagnostics. Predictions remain close to the lower end of the original-unit range while observed durations extend to roughly 20,000 minutes. The model therefore underpredicts very long-duration sites. Residuals are broadly centred but retain wide tails. The centre-compression is severe in physical units because duration is highly right-skewed.

Core finding for Model 3. Mean duration is not limited by sample size alone. The model already has 1,228 WWTWs and its learning curve is near a plateau, yet predictive skill remains weak. The report therefore points toward missing storage, hydraulic capacity, network topology, pumping/maintenance state and event-level antecedent wetness. The large company sensitivity reinforces that conclusion.

6. Model 4: predicting annualised spill frequency

Question. Can measured catchment conditions predict exposure-adjusted spill-event frequency at an unseen WWTW?

Answer. This is the strongest of the four models. Skill is still modest, but the evidence that it beats the constant baseline is clearer than for the other targets.

Data	967 total: 676 train, 145 validation, 146 test	Selected model	Ridge, full static + dynamic
Test RMSE	1.341	Dummy RMSE	1.442
Gain	7.0%	Test R^2	0.133
Spearman	0.346	Bootstrap gain	[0.8%, 12.1%]

Why this is the strongest model

- It has the largest improvement over its own dummy baseline: 7.0%.
- It has the highest test R^2 : 0.133.
- It has the strongest test-set rank association: Spearman 0.346.
- Most importantly, its bootstrap improvement interval remains above zero. Among these four models, this is the clearest evidence that the measured feature system adds out-of-sample information beyond a constant predictor.

Annual dynamic rainfall lowers test RMSE by 0.054, the largest rainfall benefit among the four targets. Company identity lowers RMSE by only 0.004, so operator identity contributes little additional held-out accuracy in this sensitivity. The most relied-on measured features are intergranular-flow geology, annual consecutive dry days and population density.

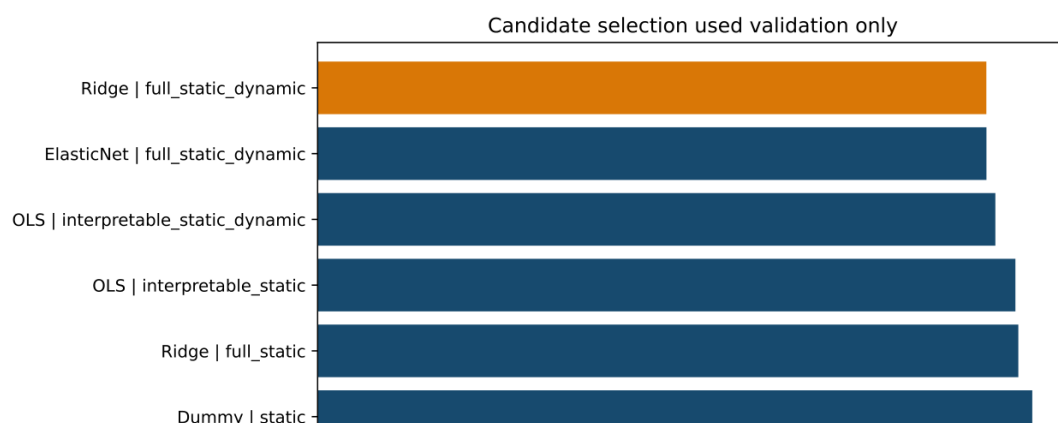


Figure 10: Validation comparison for Model 4. Ridge with full static + dynamic information is selected.

Locked-test diagnostics: strongest does not mean accurate

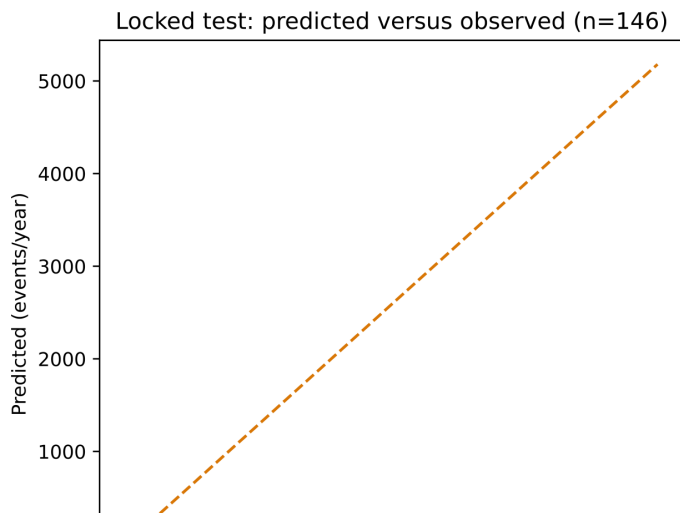


Figure 11: Observed versus predicted annualised frequency.

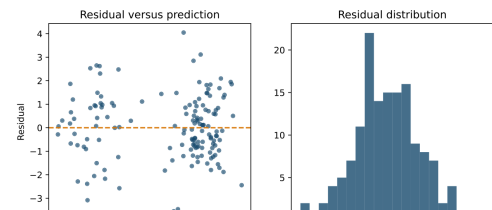


Figure 12: Residual diagnostics on the log scale.

How to read these diagnostics. The model still compresses predictions toward typical event rates and misses the highest observed frequencies. Residuals show substantial scatter around zero. Its advantage over the other models comes from doing better than its own baseline, not from reproducing the full range of site-level frequency exactly.

Core finding for Model 4. Frequency is the best current target for predictive ML using these external variables, and annual dynamic rainfall contributes most clearly here. But $R^2 = 0.133$ is still modest, so reporting/exposure variation, sewer-network state and event opportunity/rainfall alignment remain important missing information.

7. Cross-model comparison: what changes from target to target?

Target	Dynamic rainfall Δ RMSE	Company Δ RMSE	Spatial RMSE	Top measured features
β	-0.015	+0.002	0.694	Population density; intergranular-flow; low-productivity geology.
λ	+0.101	+0.014	3.776	Population density; low-productivity; intergranular-flow geology.
Duration	-0.011	-0.131	1.551	Static winter rainfall; static annual rainfall; annual SDII.
Frequency	-0.054	-0.004	1.430	Intergranular-flow geology; consecutive dry days; population density.

Sign convention. Negative Δ RMSE means the added information lowers error and therefore helps. Positive Δ RMSE means it worsens the test result. These are pre-specified sensitivity comparisons and do not redefine the frozen primary models.

Annual dynamic rainfall

It helps β , duration and frequency, but only by small amounts. The largest benefit is for frequency (-0.054). It worsens λ (+0.101). The key message is therefore not “rainfall solves the ML problem”; it is that annual rainfall supplies incremental information for some direct outcomes while λ remains unstable.

Company identity

Company is almost irrelevant for β , λ and frequency in held-out sensitivity comparisons, but substantially improves duration (-0.131 RMSE). This target-specific pattern is important because it points toward duration-relevant operator/regional processes that are not represented by the current measured variables.

Different targets reveal different information

The two fitted tail parameters are difficult partly because their labels come from a distributional fitting process. The two direct outcomes are easier to define but still contain monitoring/reporting variation and unobserved hydraulic state. The fact that the selected feature sets differ is therefore scientifically meaningful: there is no single external predictor system that explains all four targets equally well.

8. Learning curves: would simply adding more WWTWs fix the problem?

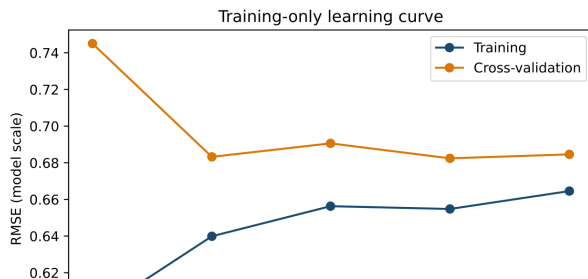


Figure 13: Training-only learning curve for β .

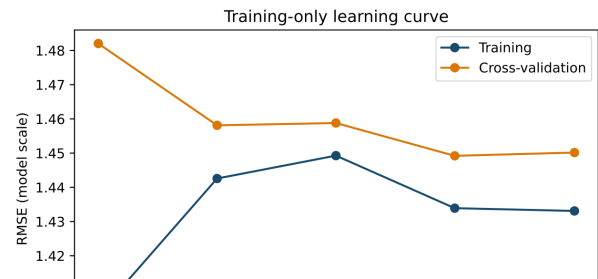


Figure 14: Training-only learning curve for frequency.

Target	n	Diagnosis from the report
β	819	Curve is near a plateau; richer predictors or improved targets may matter more than more similar rows.
λ	783	Same diagnosis.
Mean duration	1,228	Same diagnosis despite the largest cohort.
Annualised frequency	967	Strongest target, yet its curve also approaches a modest-skill plateau.

Interpretation. More WWTWs could improve precision and transfer testing, but the curves do not suggest a large performance jump from simply adding more rows with the same feature structure. The report’s priority is richer hydraulic, network and temporal information plus better target measurement.

Why a plateau matters

If cross-validation error were still falling sharply as training size increased, collecting more similar examples would be an obvious solution. Here the curves flatten. That changes the research question from “how many more rows do we need?” to “what important information is absent from each row?”

9. Why the models remain weak: limitations that actually matter

Limitation class	What the report identifies
Target quality	Fitted β/λ uncertainty; joint parameter identifiability; extreme right-skew; incomplete exposure for some operators.
Hydraulic state	No direct storage, sewer topology, pumping state, hydraulic headroom or maintenance state.
Catchment state	No direct groundwater level, soil wetness or event-specific antecedent conditions.
Rainfall coverage	Annual NIMROD indices require complete monitoring years, limiting coverage.
Operator / region	Company/reporting differences remain, particularly visible for duration.
Model form	Mainly linear/regularised relationships; collinearity; regional dependence; difficulty extrapolating to extreme outcomes.

The wrong conclusion would be: “The regressions failed, therefore we need a much more complex ML algorithm.” The report does not support that. Validation already compared OLS, Ridge and Elastic Net; the learning curves plateau; and the prediction plots show missing extreme/system-state information. More aggressive algorithms cannot recover variables that are not measured.

Target-specific improvement priorities

- **β and λ :** improve tail-fit precision, retain joint parameter covariance, add event-mixture/hydraulic descriptors, and consider hierarchical shrinkage by company before increasing algorithm complexity.
- **Duration:** add storage, hydraulic capacity, network topology, pumping/maintenance state and event-level antecedent wetness. The company sensitivity suggests major operator/region information is currently unmeasured.
- **Frequency:** improve monitored exposure, harmonise reporting, align rainfall with event opportunity, and add network/operational state. Annual dynamic rainfall is most useful for this target.

More data should prioritise broader companies/regions and higher-quality labels rather than simply increasing the number of similar WWTWs.

10. Predictive ML is not the same as inferential multivariable regression

This machine-learning programme	The forthcoming inferential analysis
Asks: can the measured feature system predict an unseen target?	Asks: which adjusted associations remain after controlling for the other measured variables?
Judged primarily by held-out RMSE, baseline improvement and transfer performance.	Judged by coefficient estimates, uncertainty, confounding control and model diagnostics.
Feature importance describes the fitted predictor system.	Coefficients describe conditional associations under an explicit model.
Does not establish causality.	Still does not establish causality automatically, but is designed for adjusted association rather than prediction.

Why this distinction matters for your supervisor. A predictor can have a small inferential p-value yet contribute little to out-of-sample prediction. Conversely, a variable can help a predictive model without representing an independent causal mechanism. The two analyses answer different questions and should be presented as complementary, not interchangeable.

What the current ML programme can legitimately claim

It can show out-of-sample predictive association, incremental information beyond a constant predictor, whether annual rainfall/company identity adds held-out accuracy, and how performance changes under spatial/operator transfer.

What it cannot claim

It cannot identify causality, prove a hydraulic mechanism, provide unbiased adjusted effect sizes, or establish that a high-importance feature directly causes spill behaviour. Correlated rainfall, geology and urbanisation features can share or redistribute importance.

11. Supervisor-facing synthesis: how to explain the results clearly

Thirty-second summary. “We trained four separate scikit-learn regression models with a strict 70/15/15 train-validation-test design. Every frozen model beats a dummy mean predictor on the locked test, but only modestly. Frequency is the strongest target, with a 7.0% RMSE improvement and $R^2 = 0.133$; its bootstrap interval is the only one entirely above zero. Beta and duration contain qualified weak signals, while lambda remains inconclusive because of fitted-parameter instability and extreme values. Annual dynamic rainfall adds small value for beta, duration and especially frequency, but not lambda. The learning curves are near plateaus, so the next gains should come from better targets and hydraulic/operational features rather than simply using a more complex algorithm.”

If asked: “Did the machine learning work?”

The precise answer is: **partially**. It worked as a rigorous test of whether the present external feature bank contains generalisable predictive information. The answer is yes, but only modestly. It did not produce a high-accuracy predictor, and the report deliberately retains that negative/qualified result rather than tuning it away.

If asked: “Which model should we trust most?”

Model 4, annualised spill frequency, has the strongest evidence because its baseline improvement is largest, its test R^2 and Spearman are highest, and its bootstrap gain interval remains positive. That still does not make it a high-skill model.

If asked: “Which model is the main problem?”

Model 2, inverse-timescale λ . Its baseline gain is uncertain, annual dynamic rainfall worsens performance, and its original-unit errors are dominated by extreme fitted values. The report therefore treats target reliability and β - λ identifiability as a priority.

If asked: “What did dynamic rainfall achieve?”

It produced incremental rather than transformative gains. It helps frequency most, helps β and duration slightly, and worsens λ . This is evidence that annual rainfall contains useful information for some targets but is not sufficient on its own.

If asked: “What do we do next?”

Preserve the locked-test results, improve target quality, add missing hydraulic/network/operational state, and preregister any new model extension against a genuinely new confirmatory sample where feasible. The forthcoming inferential multivariable regression should be kept conceptually separate from predictive ML.

Final validation status

Source masters and the historical ML directory were unchanged; WWTWs are unique with approximately 70/15/15 splits and zero overlap; test observations were excluded from selection; preprocessing remained inside pipelines; final models reload with identical predictions; RMSE was independently recalculated; and 2,000 fixed-prediction bootstrap resamples were used. The source analysis also produced 48 vector diagnostic PDFs.

Final research conclusion. Measured external conditions have modest predictive value, strongest for annualised spill frequency. Annual dynamic rainfall adds small value for β , duration and especially frequency, but not λ . Across all four targets, the dominant limitation is missing hydraulic/operational state and target quality rather than insufficient algorithm complexity.

This redesign changes presentation and explanatory depth only. Numerical results, model choices, splits, diagnostics and scientific conclusions are taken from the supplied machine-learning report; no new model fitting or external evidence has been added.